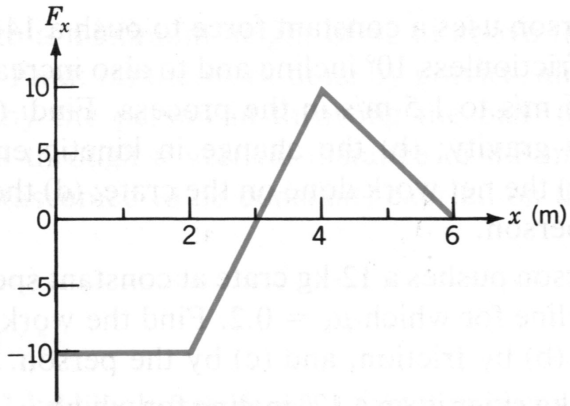
Homework #9

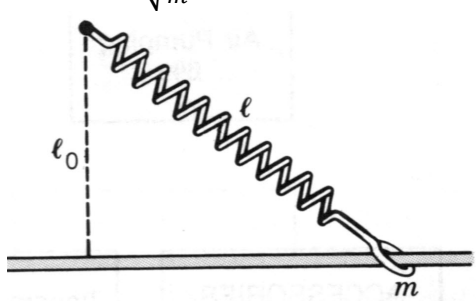
1) The force experienced by a 0.250 kg particle is depicted in the figure below. The particle approaches the origin from the right at a speed of 20.0 m/s. (a) Find the work done by the force as the particle moves from x=6.00 m to x = 0. (b) What is the kinetic energy of the particle at the origin?



2) A person lifts a 12.0 kg bucket of water from a well at constant speed. If 4000 J of work are done, how deep is the well?

3) A person uses a constant force to push a 14.0 kg crate 1.80 m up a frictionless 10° incline and to also increase its speed from 0.5 m/s to 1.5 m/s in the process. Find: (a) the work done by gravity; (b) the change in kinetic energy of the crate; (c) the net work done on the crate; (d) the work done by the person.

4) A spring of natural length l_0 and whose stiffness constant is k is fixed at one end. The other end is attached to a ring of mass m that can slide on a frictionless rod as shown in the figure below. If the initial length of the spring is l, show that the speed of the ring when the spring returns to its natural length l_0 and is perpendicular to the rod is $\sqrt{\frac{k}{m}}(l - l_0)$.



5) The only force acting on a 2.00 kg body as it moves along a positive x axis has an x- component $F_x = +6.0\,x$ N, with x in meters. The velocity at x = 3.0 m is 8.0 m/s. (a) What is the velocity of the body at x = 4.0 m? (b) At what positive value of x will the body have a velocity of 7.0 m/s?

6) In 1970 the rocket-powered car Blue Flame reached a record 1002 km/h. The thrust of its engine was 13,000 1b (58 kN). What was its maximum horsepower?

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7) Constant power P is delivered to a car of mass m by its engine. Show that if air resistance can be ignored, the distance covered in a time t by the car, starting from rest, is given by $s = \left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$. (This one is a calculus exercise, and is a little tricky. Find work in terms of velocity using work energy theorem and put into formula for average power. Then note that $v = \frac{dx}{dt}$. Finally separate variable and integrate).

8) A bungee cord exerts a nonlinear elastic force of magnitude $F(x) = k_1 x + k_2 x^3$, where x is the distance the cord is stretched, $k_1 = 204\ \mathrm{N/m}$ and $k_2 = -0.233\ \mathrm{N/m^3}$. How much work must be done on the cord to stretch it 16.7 m?